

Diagnosis of Heart Disease Prediction using Machine Learning

Milestone 1 – Design and Strategy

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Course: Big Data Analytics

1. Project Setting and Problem Statement

Heart disease is one of the top causes of death worldwide. Many people are detected late since the symptoms are not always obvious in the early stages. Hospitals and other healthcare institutions are already leveraging data analytics and machine learning to help clinicians identify patients who may be at risk.

The aim of this project is to construct a machine learning model that predicts the probability of a person having heart disease based on medical data including age, blood pressure, cholesterol level, chest pain kind and ECG results.

The major aim of the project is to support healthcare professionals, hospitals, medical researchers and patients. A good prediction system can help in early diagnosis and improved planning for therapy.

2. Choosing the Dataset

For this study I chose the Heart Disease dataset from the UCI Machine Learning Repository .

Dataset Link:

<https://archive.ics.uci.edu/dataset/45/heart+disease>

The dataset was picked since it is public, real-world, and widely utilized in healthcare machine learning research. The dataset contains the patient clinical information and can be used for disease prediction analysis.

3. Source and Background Data

The data were acquired from numerous medical institutions including the Cleveland Clinic Foundation and other hospitals engaged in cardiovascular research.

The researchers collected this data to research heart disease and to see whether clinical characteristics are associated with cardiovascular difficulties. The dataset contains patient medical and diagnostic test outcome measures.

I acquired the dataset directly from the official UCI Machine Learning Repository website.

4. Structure of a dataset

The dataset is structured in a tabular form. Some variables are:

- Age
- Sex
- Chest pain type
- Resting blood pressure

- Cholesterol level
- Fasting blood sugar
- ECG results
- Maximum heart rate achieved
- Exercise induced angina
- Number of major vessels
- Presence or absence of heart disease

The target variable indicates whether a patient has heart disease or not.

5. Data Limitations and Quality Issues

The dataset is interesting but has various limits and quality issues.

Some variables have missing values, especially “ca” and “thal” columns. These missing values will have to be pre-processed before the model is trained.

The dataset is also quite tiny compared to modern healthcare datasets, which may limit the generalizability of conclusions.

Another issue is that the data collection is old and may not accurately reflect current healthcare populations or revised standards of diagnosis.

There may also be some imbalance between patients with and without heart disease, which may impair model performance.

6. Questions for Research

The study will address the following research questions:

1. Can machine learning algorithms predict heart disease accurately using patient clinical data?
2. Which medical factors are the most relevant for heart disease prediction?
3. What is the optimal machine learning algorithm for this dataset considering assessment measures such as accuracy, precision, recall and F1-score?

7. Expected Results and Significance

An ideal outcome would be a machine learning model that can predict cardiac disease with a good accuracy and reliable performance.

This project is essential as early diagnosis of cardiac problems might assist doctors to make faster and better decisions. Knowing the major risk factors can also aid in enhancing preventative health care.

8. Privacy, Ethics and Responsible AI Review

The dataset is free of personal information including patient names, phone numbers, or addresses, therefore the privacy risk is low.

But ethical considerations remain. In cases with limited or uneven data, machine learning algorithms might generate biased or wrong predictions.

This project is intended to be used as a decision support system only and not a replacement for expert medical advice.

The dataset is publicly available for educational and research purposes via the UCI Machine Learning Repository.

9. Planned Analysis Steps

The project will follow these main steps:

1. Data collection and loading
2. Data cleaning and preprocessing
3. Exploratory Data Analysis (EDA)
4. Feature engineering and encoding
5. Model building using machine learning algorithms
6. Model evaluation using accuracy, precision, recall, F1-score, and ROC-AUC
7. Interpretation of results and comparison of models

10. Tools and Technologies

The following tools will be used during the project:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib and Seaborn
- Scikit-learn
- GitHub

11. Statement on the Use of AI

All material, code and explanations were checked and confirmed before being added to the project. No AI tools will be used to generate false results, and no AI tools.

12. References

1. UCI Machine Learning Repository – Heart Disease Dataset

<https://archive.ics.uci.edu/dataset/45/heart+disease>

2. Scikit-learn Documentation

<https://scikit-learn.org/stable/>

3. Pandas Documentation

<https://pandas.pydata.org/docs/>

13. Suggested File Name

KaishivJoshi_Milestone1_CIND820_Big Data Analytics Project.pdf